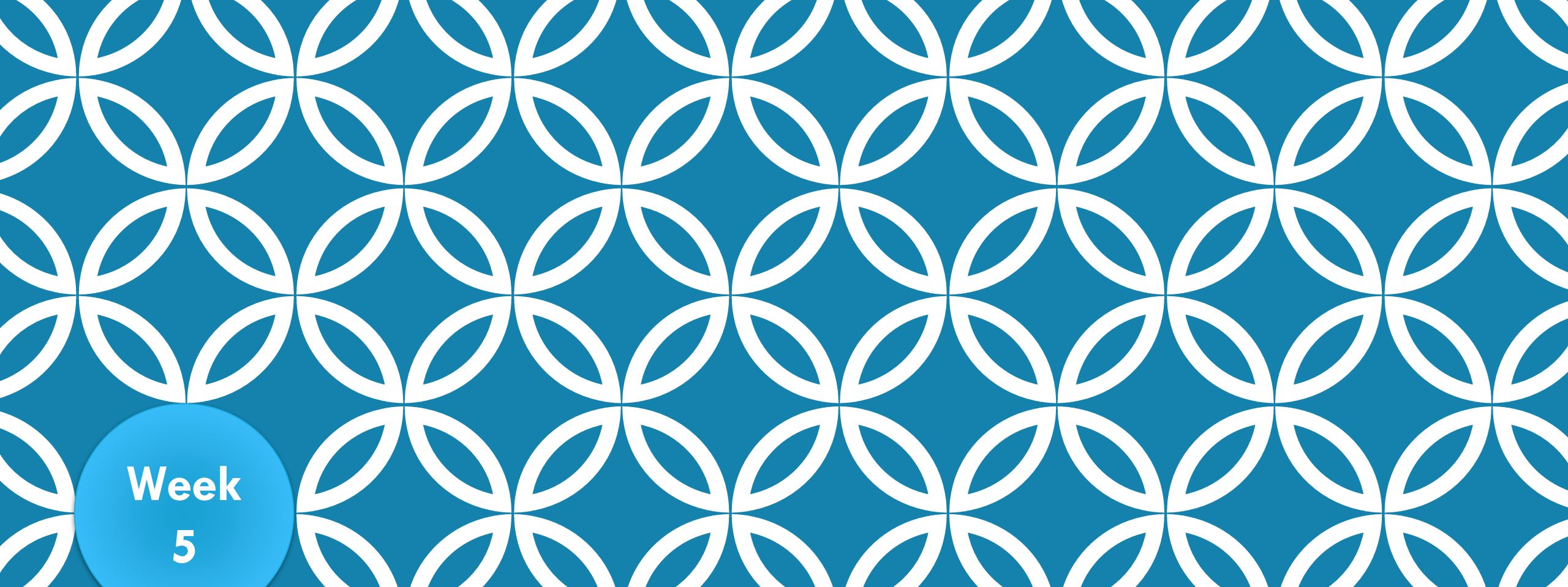




**Week
5**

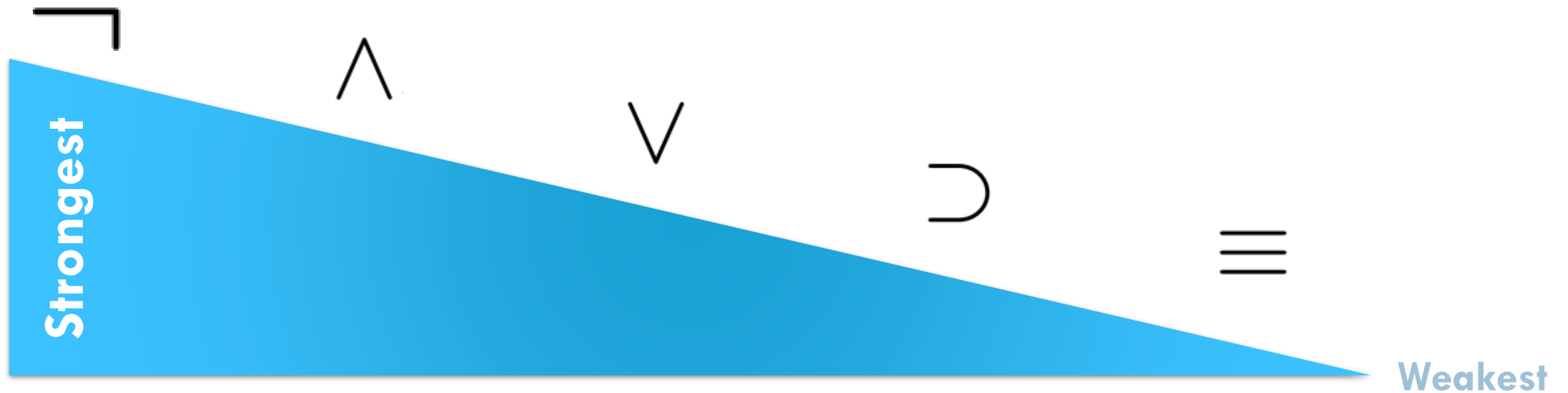
1. Logical consequences

2. Equivalency

Logic in Computer Science Lab
Gyöngyi Bujdosó
2025

Based on Tamás Kádek's lectures

Precedence



Parentheses can be dropped when the structure of an expression can be determined by precedence.

Precedence

If surrounded by two occurrences of $\wedge$ or $\vee$, the operand associates with the operator to the left.

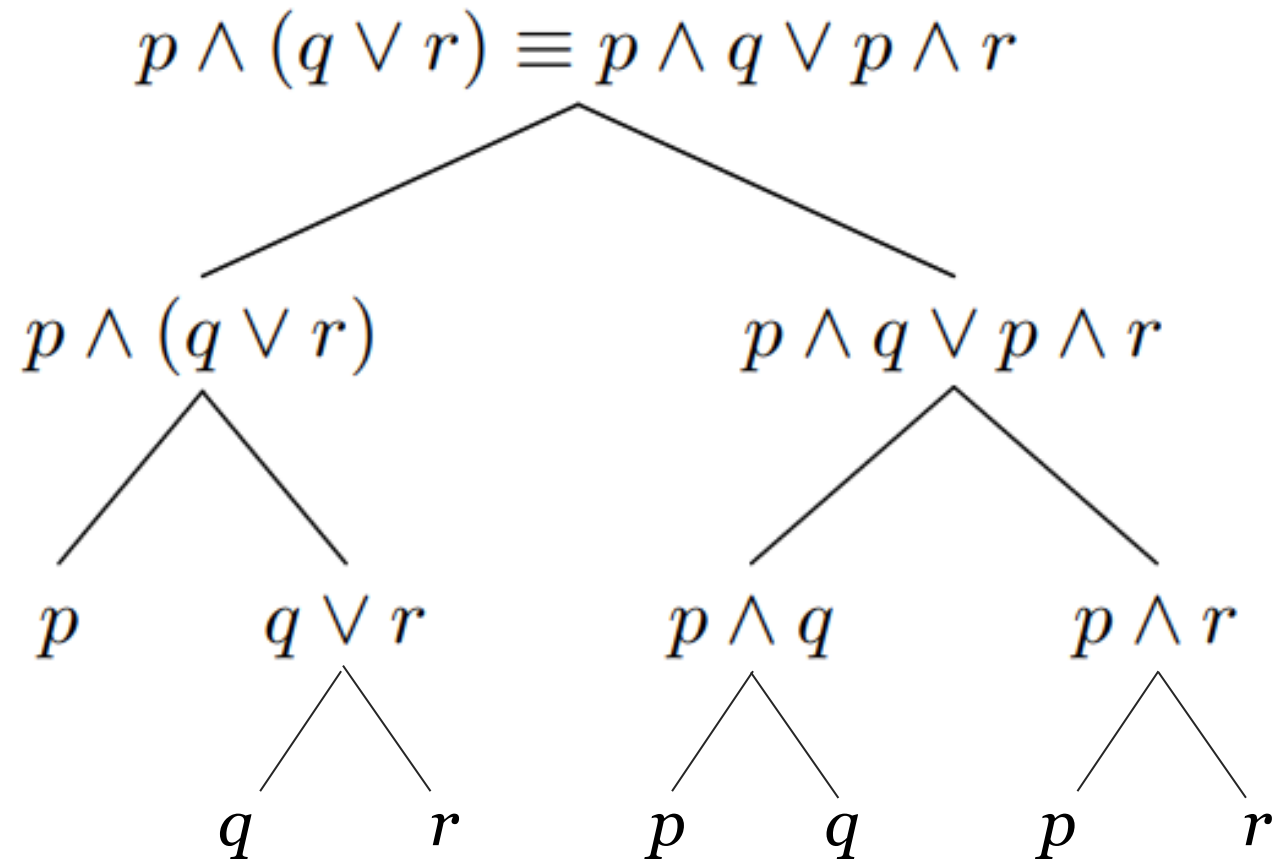
$$\begin{aligned} p \wedge q \wedge r &\rightarrow ((p \wedge q) \wedge r) \\ p \vee q \vee r &\rightarrow ((p \vee q) \vee r) \end{aligned}$$

If surrounded by two occurrences of $\supset$ or $\equiv$, the operand associates with the operator to the right.

$$\begin{aligned} p \supset q \supset r &\rightarrow (p \supset (q \supset r)) \\ p \equiv q \equiv r &\rightarrow (p \equiv (q \equiv r)) \end{aligned}$$

Precedence

$$p \wedge (q \vee r) \equiv p \wedge q \vee p \wedge r$$



2-argument Boolean functions: $f : \{0, 1\}^2 \rightarrow \{0, 1\}$

2-argument dependency

Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
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	$\equiv$	$\neq$	$\supset$	$\vee$	$\parallel$	$\wedge$	$ $
0 0	1	0	1	0	1	0	1
0 1	0	1	1	1	0	0	1
1 0	0	1	0	1	0	0	1
1 1	1	0	1	1	0	1	0



XOR



Peirce's arrow



Sheffer stroke

1-argument dependency

Identity 1	Negation 1	Identity 2	Negation 2
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π_1	π'_1	π_2	π'_2
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0

No argument dependency

Falsity	Truth
---------	-------

$\perp$	$\top$
0	1
0	1
0	1
0	1

Law of the double negation:

$$\neg\neg A \Leftrightarrow A$$

Models, satisfiability

Definition (model of a set of formulas)

Let $\Gamma \subseteq \text{Form}$ be a set of formulas. An interpretation ϱ is a **model of the set of formulae Γ** , if $|A|_{\varrho} = 1$ for all $A \in \Gamma$.

Definition (satisfiable - set of formulas)

The **set of formulae $\Gamma \subseteq \text{Form}$ is satisfiable** if it has a model.

- If there is an interpretation in which all members of the set Γ are true.

Definition (unsatisfiable set)

The **set $\Gamma \subseteq \text{Form}$ is unsatisfiable** if it is not satisfiable.

Logical consequence

Definition

A formula A is the logical consequence of the set of formulae Γ if the set $\Gamma \cup \{\neg A\}$ is unsatisfiable. (Notation : $\Gamma \models A$)

Theorem

Let $\Gamma \subseteq Form$, and $A \in Form$. $\Gamma \models A$ if and only if all models of the set Γ are models of formula A . (i.e. the singleton $\{A\}$).

Corollary

Let $\Gamma \subseteq Form$, and $A \in Form$. $\Gamma \models A$ if and only if for all interpretations in which all members of Γ are true, the formula A is true.

The notation

$$A \models B$$

(B is a logical consequence of A)

means that

B is true

in every interpretation

in which

A is true.



**In those cases
when the assumptions are true,
the conclusion is true**

**In those cases
when the conclusion is false
at least 1 assumption is false**

Validity

Definition

The **formula A is valid** if $\emptyset \models A$. (Notation: $\models A$)

Logical equivalence

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language and $A \in Form$ and $B \in Form$ be two formulas of $\mathcal{L}^{(0)}$.

The formula A and formula B are *logically equivalent*, if

$$A \models B \quad \text{and} \quad B \models A.$$

(Notation: $A \Leftrightarrow B$)

Logical equivalence — Example

We have two formulae: $(\neg p \vee q)$ and $(p \supset q)$

Are they logically equivalent?

Tautology — Contingency — Contradiction

Tautology

p	$p \vee \neg p$
0	1
1	1

Satisfiable

Model 1: $|p|_e = 0$

Model 2: $|p|_e = 1$

Contingent formula

p	q	$((p \wedge \neg p) \vee q)$
0	0	0
0	1	1
1	0	0
1	1	1

Satisfiable

Model 1: $|p|_e = 0$ and $|q|_e = 1$

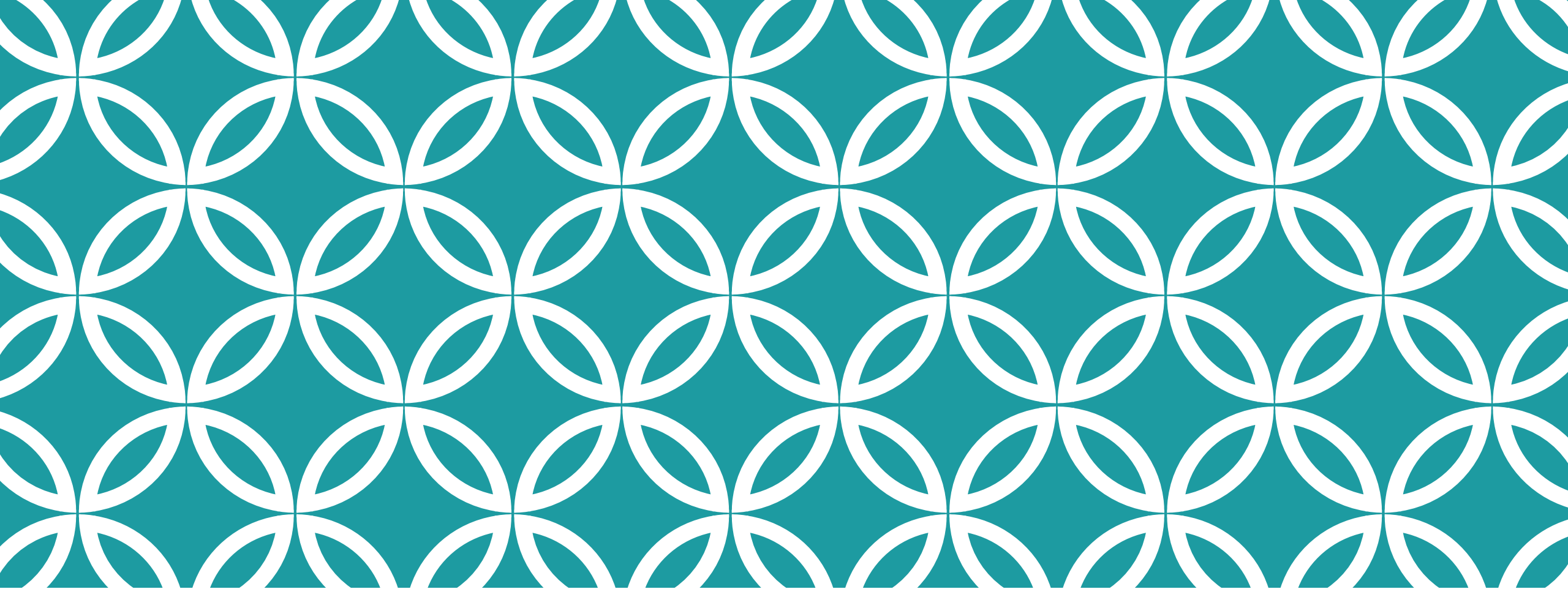
Model 2: $|p|_e = 1$ and $|q|_e = 1$

Contradiction

p	$p \wedge \neg p$
0	0
1	0

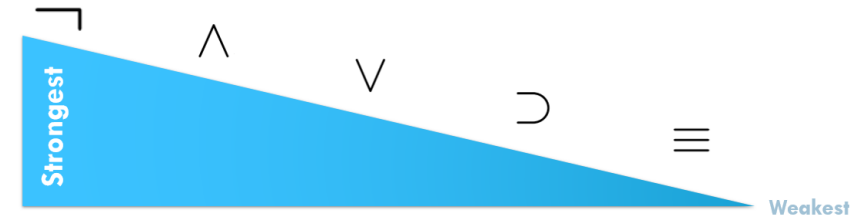
Unsatisfiable /
Not satisfiable

It has no model



Exercises

Logical consequences



17. Which formulae are logical consequences of the formula $\neg(q \wedge p)$?
☐ $\neg(q \vee \neg q)$ ☐ $\neg(p \vee \neg q)$ ☐ $(q \equiv \neg p)$ ☐ $((q \equiv p) \supset p)$ ☐ $(q \supset \neg p)$
18. Which formulae are logical consequences of the formula $(q \supset \neg p)$?
☐ $\neg(p \vee q)$ ☐ $(p \supset \neg q)$ ☐ $((q \wedge p) \wedge q)$ ☐ $(q \supset (q \supset p))$ ☐ $(q \vee (p \equiv p))$
19. Which formulae are logical consequences of the formula $(\neg p \vee q)$?
☐ $\neg(p \wedge q)$ ☐ $(p \supset (q \wedge p))$ ☐ $(q \wedge (q \equiv p))$ ☐ $(\neg p \supset p)$ ☐ $(q \equiv (p \supset q))$
20. Which formulae are logical consequences of the formula $(q \equiv (q \wedge p))$?
☐ $\neg(p \supset \neg p)$ ☐ $((q \vee p) \supset p)$ ☐ $(q \vee (q \equiv p))$ ☐ $(q \wedge (p \supset p))$ ☐ $(q \equiv (p \wedge p))$
21. Which formula sets have the logical consequence of formula $\neg(q \equiv \neg q)$?
☐ $\{(q \vee (q \vee p)), (p \equiv \neg \neg q)\}$
☐ $\{((q \vee p) \equiv p), (q \wedge (q \supset p))\}$
☐ $\{((p \supset p) \supset p), (q \wedge p)\}$
☐ $\{((p \wedge p) \wedge p), \neg(p \wedge p)\}$
☐ $\{(p \supset (p \equiv q)), ((q \vee p) \wedge q)\}$

Logical consequence

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\parallel$	$\wedge$	$ $
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

22. Which formula sets have the logical consequence of formula $\neg(p \vee \neg p)$?

- ☐ $\{((p \wedge p) \equiv q), \neg(q \vee p), \}$
- ☐ $\{(p \equiv \neg p), \neg q \wedge q\}$
- ☐ $\{((p \equiv q) \wedge p), \neg(p \wedge p)\}$
- ☐ $\{\neg(\neg q \supset q), ((p \wedge q) \vee p)\}$
- ☐ $\{(q \equiv \neg p), ((q \vee p) \wedge q)\}$

23. Which formula sets have the logical consequence of formula $\neg(q \vee p)$?

- ☐ $\{(q \supset \neg p), (\neg q \supset \neg p)\}$
- ☐ $\{(p \equiv (p \supset q)), (p \vee \neg q)\}$
- ☐ $\{\neg(p \vee p), (\neg q \vee \neg p)\}$
- ☐ $\{\neg(\neg p \supset p), (q \supset (p \wedge q))\}$
- ☐ $\{\neg q, (\neg p \supset p)\}$

Logical consequence

		Equivalent	Not equivalent	Implies	OR Disjunction	NOR	AND Conjunction	NAND
		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\parallel$	$\wedge$	$ $
0	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	1	1	0	1	0

- Is the logical consequence of formulae p and $(p \supset q)$ the formula q ? Justify your answer!
- Is the logical consequence of formulae $(p \supset q)$ and $(q \supset r)$ the formula $(p \supset r)$? Justify your answer!
- Is the logical consequence of formulae $(p \vee (q \vee r))$, $\neg q$ and $\neg r$ the formula p ? Justify your answer!
- Is the logical consequence of formulae $(p \supset q)$ and $\neg q$ the formula $\neg p$? Justify your answer!
- Is the logical consequence of formulae $(p \supset (q \supset r))$ and q the formula r ? Justify your answer!

Logical equivalency

46. The formulae $((p \vee q) \wedge r)$ and $((p \wedge r) \vee (q \wedge r))$ are logically equivalent? Justify your answer!
47. The formulae $(p \supset q)$ and $(q \supset p)$ are logically equivalent? Justify your answer!
48. Which of the following formulae are logically equivalent to formula $(\neg p \wedge p)$?
☐ $((q \supset p) \wedge q)$ ☐ $(q \wedge \neg q)$ ☐ $(\neg p \wedge \neg q)$ ☐ $\neg(p \supset q)$ ☐ $\neg(q \vee \neg q)$
49. Which of the following formulae are logically equivalent to formula $\neg(p \vee q)$?
☐ $\neg(\neg p \supset q)$ ☐ $(p \wedge \neg q)$ ☐ $(q \wedge (p \equiv q))$ ☐ $(q \wedge \neg p)$ ☐ $\neg(p \vee p)$
50. Which of the following formulae are logically equivalent to formula $(\neg p \wedge q)$?
☐ $\neg(\neg q \vee p)$ ☐ $(q \wedge (q \supset p))$ ☐ $\neg(q \supset p)$ ☐ $\neg(q \vee p)$ ☐ $\neg(q \vee \neg p)$
51. Which of the following formulae are logically equivalent to formula $\neg(\neg p \supset p)$?
☐ $\neg(q \equiv p)$ ☐ $\neg p$ ☐ $((p \supset q) \supset p)$ ☐ $(p \equiv (q \vee q))$ ☐ $(q \supset \neg q)$